CONTACT

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Location Jersey City, NJ

EDUCATION

M.S. Biomedical Sciences
Auburn University College of Veterinary Medicine · 2020

B.S. Biomedical Sciences
Minor: Biomedical Physics
University of South Florida · 2016

MOLECULAR & PCR ASSAYS

- qPCR / RT-qPCR / dPCR-ready workflows
- Assay development, optimization & qualification
- Viral titer & genomic-copy quantification (10^3 to 10^{12} range)
- Residual-DNA & contamination QC concepts
- Primer design & amplification strategy
- Sanger & NGS construct verification

GENE EDITING & VECTORS

- CRISPR/Cas9 gene editing
- AAV, lentivirus, oncolytic VSV
- Gibson, Golden Gate & seamless cloning
- Traditional molecular cloning
- Site-directed mutagenesis
- Construct design → rescue → titration
- Full-genome assembly (11,161 bp VSV)
- Glycoprotein retargeting & aptazyme switches

CELL & IMMUNOASSAYS

- Multicolor flow cytometry, FACS & MACS
- Mammalian cell & tissue culture (13+ lines)
- Cell factories & scaled adherent culture
- Single-cell isolation & rare-cell sorting
- Cell-based & immune-cell assays
- ELISA, Western blot, SDS-PAGE
- TCID50 & plaque assay (viral titration)
- Virus production, purification & titration
- Targeted protein neutralization
- Ultrafiltration/diafiltration (UF/DF)
- Density-gradient purification

GENOMICS & NGS

- Illumina, Oxford Nanopore, PacBio
- DNA/RNA library prep, QC & methylation
- Multiplexed / barcoded library workflows
- Direct RNA-seq (SQK-RNA004 chemistry)
- Bisulfite conversion & epigenomics
- Base-calling, alignment & variant ID
- Single-cell sequencing pipelines
- FFPE nucleic-acid extraction & prep

INSTRUMENTS & SOFTWARE

- Qubit, TapeStation, qPCR (QC)
- BioMek, Dragonfly, FAST (automation)
- Incucyte live-cell imaging
- HPLC & fluorescence microscopy
- IGV & Galaxy (sequence analysis)
- ELN; SOP & test-method authoring
- GraphPad Prism; MS Office & Google Workspace

CERTIFICATIONS

TAYLOR FLAAT

Senior Research Scientist

Gene Editing & Molecular Assays · Method Development & Qualification · Viral Vectors

PROFESSIONAL SUMMARY

Research Scientist with ten years of hands-on molecular-biology experience across academia, biotech startups, and pharma, with deep expertise in developing, optimizing, and qualifying molecular assays for gene- and cell-therapy modalities. Built and validated PCR-based quantification methods (qPCR, RT-qPCR) for viral titer and genomic-copy measurement, designed and characterized AAV, lentiviral, and oncolytic VSV vectors end-to-end, and executed rigorous research-grade QC using Qubit, TapeStation, and qPCR. Skilled across synthetic virology, oncolytic virus therapy, precision oncology, gene editing, and high-throughput NGS, with a proven ability to translate complex biological problems into actionable research strategies. Experienced authoring SOPs and test methods, maintaining contemporaneous lab-notebook records, and mentoring junior scientists. GLP/GCP-trained and eager to apply established method-development rigor within a cGMP analytical-development environment supporting first-in-human and IND products.

KEY PROJECTS & MILESTONES

Synthetic VSV Genome — 11,161 bp Assembled From Four Modular Fragments
Synthetic Virology · Humane Genomics · Peer-reviewed in Viruses 2024, 16(10):1641

- Co-developed a synthetic-biology platform to engineer Vesicular Stomatitis Virus from the ground up: a full-length **11,161 bp genome assembled from four modularized DNA fragments**, rescued and titered, with phenotypic analysis showing no significant difference between natural and synthetic virus.
- Demonstrated platform flexibility by swapping in a foreign glycoprotein and rearranging gene order (VSV-P ⇄ VSV-M) to prove design freedom; hundreds of engineered virions sequenced by Nanopore for construct verification.

VSV-Vectored COVID-19 Vaccine Candidate — Bench to In Vivo in Five Months
Vaccine Development · Pandemic Response · Humane Genomics

- Built a SARS-CoV-2 vaccine candidate on the same VSV backbone behind Merck's approved Ebola vaccine, swapping the native glycoprotein for the **SARS-CoV-2 spike protein**; designs went from concept to functional virus particles in under one month.
- Confirmed surface display of spike and **ACE2-receptor binding** by Incucyte live-cell imaging; scaled production and purification **100-1000x** to supply animal studies.
- Supported a rat safety study at Auburn's Scott-Ritchey Research Center (well tolerated, no adverse reactions; CBC and clinical chemistry confirmed), contributing to a **pre-IND submission to the FDA** before the program was wound down once authorized vaccines reached the public.

Rapid Viral Screening Pipeline — 24 Genomes, Supernatant to Sequence in 24 Hours
Method Development · Nanopore Sequencing · Humane Genomics

- Built a single end-to-end workflow taking a candidate virus from harvested supernatant to called sequence in one day: supernatant harvest → viral RNA isolation → cDNA synthesis → PCR amplification → library prep on the Oxford Nanopore **Rapid Barcoding Kit** → sequencing → assembly and analysis.
- Barcoding multiplexed **up to 24 viral genomes in a single run**, converting a serial, multi-day sequencing task into a parallel screen and delivering a same-day readout fast enough to steer the next round of engineering.

Two-Step RT-qPCR Method for Viral Titer Quantification
Assay Development · qPCR · Humane Genomics

- When low RNA yield and purity from viral passaging broke one-step RT-qPCR, engineered a **two-step RT-qPCR method** that cleanly quantified viral genomes across a **10^3 to 10^{12} genome-copy dynamic range**.
- This titer data dosed the in vivo mouse studies and seeded the lab's genome-sequencing workflow; the method is directly analogous to VCN, genomic-titer, and infectious-titer assays used in cell- and gene-therapy QC.

Retargeted Oncolytic VSV With a Genetic On-Switch for Liver Cancer
Oncolytic Therapy · AACR Annual Meeting 2023 · Humane Genomics

- Google AI Professional Certificate (2026)
- GLP / GCP Training

HIGHLIGHTS

- Startup-to-big-pharma experience
- Engineered 11,161 bp synthetic viral genome
- Built qPCR & NGS pipelines from scratch
- COVID-19 vaccine candidate to pre-IND
- 24-plex viral genome screen in 24 hours
- First published canine melanocyte protocol
- AI for data & discovery

- Engineered oncolytic VSV to selectively kill **GPC3-positive (glypican-3)** liver-cancer cells, pairing a retargeted glycoprotein with **aptazyme replication switches**.
- Demonstrated infection and lysis at low MOI with no effect in GPC3-negative cells; 10^7 PFU doses well tolerated in vivo.

Two-Hour Nanopore Plasmid-Sequencing Method

Method Development · Nanopore · Humane Genomics

- Built an indirect viral-sequencing route (RNA → cDNA → dsDNA → Nanopore) to sequence engineered virions base-by-base for QC, then compressed the plasmid workflow to **miniprep-to-analysis in two hours**, with assembly and analysis in Galaxy.

Rare-Cell Isolation Protocols for Precision Oncology

Precision Oncology · M.S. Thesis · Auburn University CVM

- Built protocols to isolate the normal-cell counterparts of three skin tumors (melanocytes, keratinocytes, and mast-cell progenitors) as transcriptomic comparators for precision oncology.
- Delivered the **first published method for canine melanocyte isolation** from skin and oral mucosa, a rare-cell target making up just 3-5% of skin cells; the protocol has since been independently reproduced by other labs.
- Isolated circulating mast-cell progenitors (~0.1-0.5% of nucleated blood cells) by MACS $CD90^+$ depletion followed by FACS of the $CD117^+$ $Fc\varepsilon RI^+$ $CD90^-$ phenotype, yielding single-cell-sequencing-grade material.

Multiplexed Direct RNA Sequencing — Service Concept (In Development, 2025-2026)

Research & Service Concept · Oxford Nanopore SQK-RNA004

- Developing a service concept to scale native RNA-seq from single-sample pilots to **96-plex cohort studies** on Oxford Nanopore SQK-RNA004 chemistry, sequencing the true RNA strand with no cDNA or PCR bias.
- Pooling a full well-plate onto a single PromethION run collapses per-sample cost from roughly **$900 to about $10 (≈99% reduction)** while preserving direct-strand sensing for m6A, pseudouridine, isoform, and poly(A) analysis; demultiplexing via SeqTagger.

Biotech Career Hub — Self-Built Web Application

Software · Applied AI Tooling · Live at biotech-career-hub-56721929391.us-west1.run.app

- Designed and shipped a live web application for navigating biotech careers, built hands-on with modern AI tooling, applying the skillset behind the Google AI Professional Certificate.

Portfolio Website — taylorflaat.com

Software · Self-built, self-contained single-page application

- Designed, vibe-coded, and deployed a scientific portfolio site presenting protocols, project case studies, and publications, hosted on GitHub Pages with a custom domain.

CAREER HIGHLIGHTS

- 10 years spanning academia to early-stage startups to big pharma
- Founding scientist: built an entire startup lab from day one
- Engineered an 11,161 bp synthetic viral genome from modular fragments
- Produced a COVID-19 vaccine candidate from in vitro through in vivo trials and pre-IND
- Built and validated qPCR and NGS pipelines from the ground up
- Developed a 24-plex Nanopore screen taking viral supernatant to sequence in 24 hours
- First published protocol for canine melanocyte isolation, independently reproduced by other labs
- Upstream/downstream bioprocessing incl. UF/DF, density-gradient viral purification & QC
- 6+ years of multicolor flow cytometry / FACS isolating rare cells
- High-throughput screening and NGS across Illumina, Nanopore & PacBio
- Viral titration by TCID50, plaque assay, and qPCR across a 10^3-10^{12} range
- ELISA, Western blot & SDS-PAGE for protein characterization
- Authored SOPs and test methods enabling lab scale-up
- Applied AI tooling for data analysis and discovery (Google AI cert.); shipped two live web apps
- Biomedical Physics minor: HPLC, fluorescence microscopy & instrumentation
- Peer-reviewed publications, conference posters & patents

INDUSTRY EXPERIENCE

Scientist I — **Sept 2024 – Mar 2026**

Azenta Life Sciences — CRO for next-generation sequencing applications

- Developed, optimized, and qualified custom DNA/RNA NGS library-prep workflows across Illumina, Oxford Nanopore, and PacBio, improving throughput and reproducibility across high-volume projects under standardized protocols, including CRISPR-validation, targeted-genomics, and whole-genome applications.
- Executed a wide range of molecular-biology techniques (PCR, reverse transcription, fragmentation, adapter ligation, and bead-based purification) for high-quality library generation.
- Maintained rigorous, standardized QC via Qubit, TapeStation automated electrophoresis, and qPCR; performed data review and verification on high-volume sample sets and troubleshot degraded FFPE inputs to deliver sequenceable libraries.
- Operated and maintained BioMek, Dragonfly, and FAST automation for high-throughput library pooling and preparation; managed equipment, calibration, and lab inventory.
- Executed whole-genome methylation library prep (bisulfite conversion, methylation-specific adapter ligation) for epigenomic profiling.
- Mentored and trained junior scientists and interns on library-prep methods, automation, and molecular-biology best practices, promoting a culture of collaboration and knowledge-sharing.
- Managed multiple parallel projects under tight timelines with consistent attention to detail and reproducibility.

Scientist II — Associate (Contract) — **Feb 2024 – Sept 2024**

Merck via On-Board Companies — Rahway, NJ · Sanger & next-generation sequencing, library preparation

- Executed advanced molecular cloning (Gibson Assembly, Golden Gate Assembly, site-directed mutagenesis) and DNA library building supporting R&D in genetic engineering and synthetic biology within a large-pharma environment.
- Performed Nanopore, Ion Torrent, and Sanger sequencing for genetic validation, genome assemblies, and construct-integrity confirmation; carried out DNA/RNA extraction, plasmid purification, transformation, electroporation, and colony PCR for gene-of-interest investigation.
- Performed microbiology and bacterial transformation/culture to facilitate genetic-engineering and microbiological studies.
- Managed and optimized laboratory protocols to enhance efficiency and accuracy across experimental workflows.
- Analyzed and interpreted experimental data, providing detailed reports and recommendations to inform project decisions.
- Collaborated across internal and external teams to drive projects from experimental design through data delivery, as well as inventory ordering and management.

Research Associate II — **Dec 2021 – Oct 2023**

Humane Genomics — New York, NY · Synthetic-virology startup engineering viruses for precision cancer therapy

- Advanced a **VSV-vectored COVID-19 vaccine candidate** from construct design through in vivo safety studies: displayed SARS-CoV-2 spike on a VSV backbone, confirmed ACE2-receptor binding by live-cell imaging, scaled production and purification 100-1000x for animal work, and supported a rat safety study and a pre-IND submission to the FDA.
- Independently developed and optimized a **real-time qPCR assay for viral quantification** across a 10^3-10^{12} genome-copy range, improving precision and reproducibility of genomic-titer and efficacy measurements, directly analogous to VCN, genomic-titer, and infectious-titer methods used in cell- and gene-therapy QC.
- Designed, built, and validated engineered viral vectors (AAV, lentivirus, oncolytic VSV) from construct design through functional validation, applying an advanced gene-editing toolkit (Gibson & Golden Gate Assembly, site-directed mutagenesis, CRISPR/Cas9, PCR/qPCR/RT-qPCR).
- Engineered **retargeted oncolytic VSV with aptazyme replication switches** for GPC3-positive liver cancer, demonstrating selective lysis at low MOI in vitro and in vivo tolerability of 10^7 PFU doses (presented at AACR 2023).
- Built a **24-plex Nanopore screening pipeline** taking viral supernatant to called sequence within 24 hours (viral RNA isolation → cDNA → PCR → Rapid Barcoding Kit → assembly), turning a serial multi-day task into a parallel screen; also compressed plasmid QC sequencing to a two-hour miniprep-to-analysis method.

- Scaled upstream production in multi-layer cell factories and purified viral stocks by ultrafiltration/diafiltration (UF/DF) and density-gradient methods; ran ELISA and Western blot for titer, purity, and protein-expression QC.
- Produced, purified, and titered viral stocks (AAV, lentivirus, VSV) underpinning all downstream in vitro and in vivo studies, establishing the quantitative release-style measurements for each lot.
- Built multicolor flow cytometry / FACS and cell-based assays to characterize transduction efficiency, target engagement, viability, and immune populations across cancer and immune-cell models.
- Conducted immunological assays supporting immunotherapy studies for cancer and autoimmune diseases, including validation of new targets and characterization of therapeutic agents.
- Developed a downstream bioprocessing project in preparation for clinical trials; authored test methods, SOPs, and downstream workflows and maintained contemporaneous records.
- Managed studies across 13 active mammalian cell lines; trained and mentored new researchers on assay execution and experimental design.

Molecular-Biology Technique Depth

- DNA/RNA extraction by organic, non-organic, and adsorption methods, ensuring high-quality nucleic-acid samples for downstream applications.
- PCR and rolling circle amplification with a well-honed primer-design skillset; gel electrophoresis for size separation and analysis of nucleic acids.
- Molecular cloning across traditional, seamless, Gibson, HIFI, and Golden Gate Assembly workflows, independently designing and executing cloning projects end-to-end.
- Site-directed mutagenesis for targeted genetic modification; bacterial transformation and culture, maintaining strains such as E. coli.

Sequencing & Data Analysis (Sanger / Nanopore)

- Hands-on Sanger and Nanopore sequencing with accurate, reliable results across iterative design cycles.
- Sequencing data analysis including base-calling, sequence alignment, and identification of genetic variations using IGV and Galaxy.

Research Associate I → Research Technician — Nov 2019 – Dec 2021

Humane Genomics — New York, NY · Founding bench scientist

- Founding laboratory scientist, one of three people at the company, single-handedly built and operated all bench-side research and QC functions.
- Performed molecular cloning, DNA/RNA extraction, transformation, electroporation, and mammalian cell culture to translate early-stage designs into reproducible, verified engineered constructs.
- Contributed Nanopore sequencing and viral-genome assembly for construct verification across iterative design cycles; authored foundational SOPs that enabled the company to scale.
- Supported early lab operations and workflow build-out, including experimental-workflow design and resource planning.

ACADEMIC RESEARCH

Graduate Research Assistant — 2018 – 2020

Auburn University College of Veterinary Medicine — Pathobiology · Dr. Bruce Smith

- Developed novel rare-cell isolation protocols (MACS $CD90^+$ depletion / $CD117^+$ enrichment followed by FACS), overcoming purity, yield, and sequencing-compatibility challenges to deliver single-cell-sequencing-grade material for precision oncology.
- Delivered the first published method for isolating melanocytes from canine skin and oral mucosa via differential adhesion in phorbol-ester-free Melanocyte Growth Medium M2, a protocol since independently reproduced by other labs.
- Ran a precision-oncology project targeting rare cell populations in blood and tissue using FACS, multicolor flow cytometry, and single-cell sequencing to identify targets for patient treatments.
- Authored a Master’s thesis and contributed to posters and publications at Auburn’s Scott-Ritchey Research Center.

Graduate Research Assistant — 2016 – 2018

Auburn University College of Veterinary Medicine — Dr. Tatiana Samoylova Lab

- Supported a companion-animal contraceptive program using Gibson Assembly, site-directed mutagenesis, PCR, RT-PCR, qPCR, and cell culture alongside graduate coursework in biomedical sciences.

Undergraduate Laboratory Research **Fall 2015**

University of South Florida — Tampa, FL · Dr. Mary Jones-Mason Lab

- Studied effects of aspartame on tumor models via in vitro cell-based assays and in vivo mouse models; performed gel electrophoresis, PCR, and cloning for molecular characterization.

PUBLICATIONS & POSTERS

Leveraging Synthetic Virology for the Rapid Engineering of Vesicular Stomatitis Virus (VSV)

Moles CM, Basu R, Weijmarshausen P, Ho B, Farhat M, Flaat T, Smith BF — Viruses 2024, 16(10):1641 · doi:10.3390/v16101641 · Peer-reviewed, open access

Abstract. Vesicular stomatitis virus (VSV) is a prototype RNA virus instrumental in advancing understanding of viral molecular biology, with applications in vaccine development, cancer therapy, and antiviral screening. This work reports the design and construction of a full-length synthetic VSV genome (11,161 bp) assembled from four modularized DNA fragments. Following rescue and titration, phenotypic analysis showed no significant differences between natural and synthetic viruses. VSV was engineered with a foreign glycoprotein to demonstrate the platform's utility, and its genome was modified by rearranging gene order (switching the positions of VSV-P and VSV-M) to exemplify design freedom. This represents a significant technical advance, providing a flexible, cost-efficient platform for rapid VSV genome construction and facilitating the development of innovative therapies.

Oncolytic VSV with Retargeted Glycoprotein and Genetic On-Switch for Precise Targeting of Liver Cancer

Moles CM, Basu R, Flaat T, Ho B, Farhat M, Weijmarshausen P, Smith BF, Whitt M — Poster, AACR Annual Meeting, April 2023

Abstract. Hepatoblastoma and hepatocellular carcinoma are common pediatric liver tumors with low survival rates and no currently effective therapies. Oncolytic virus therapy is a promising modality that leverages engineered viruses to replicate in and kill cancer cells selectively. We developed a synthetic-biology platform to engineer oncolytic VSV therapies with high specificity and efficacy for liver cancer; VSV was selected for its high lytic potential, retargetability, low seroprevalence, and well-characterized small genome. Engineering goals included developing liver-cancer-specific glycoprotein and genetic replication-switch components, determining selectivity and efficacy in vitro, and evaluating safety and biodistribution in vivo. Glycoproteins were designed to target glypican-3 (GPC3), a cell-surface proteoglycan used as a diagnostic biomarker; aptazymes, composed of a ribozyme and a cancer-specific aptamer, served as genetic replication switches. In vitro testing showed infection and lysis in target cells at low MOI, with no effect in GPC3-negative cells; in vivo safety studies showed that 10^7 PFU doses were well tolerated. Results demonstrate that rationally designed oncolytic viruses can be engineered efficiently with high specificity and efficacy, showing strong promise for therapeutic development.

Methods for Isolating Canine Mast Cell Progenitors, Melanocytes, and Keratinocytes

Flaat TD — Master's Thesis, Auburn University College of Veterinary Medicine, April 2020 · etd.auburn.edu/handle/10415/7140

Abstract. One of the most difficult aspects of employing precision medicine in cancer treatment is determining the best targets within the tumor. A novel approach is to sequence the tumor transcriptome to identify genes whose expression or regulation are altered, which requires normal-cell transcriptomes as a comparator. In this study, protocols were developed to isolate normal cells from canine tissue and blood corresponding to three tumor types: melanoma, mast cell tumor, and squamous cell carcinoma. Epidermal melanocytes (~3-7% of skin cells) were enriched from oral mucosa; circulating mast cell progenitors (~0.1-0.5% of blood cells) were isolated using MACS depletion of $CD90^+$ cells followed by FACS isolation of $CD117^+$ cells; keratinocytes were successfully isolated with minimal melanocyte contamination. Protocols were successfully developed for both mast cell progenitors and keratinocytes, yielding usable sequencing data for precision oncology studies.

Identification of Canine Mast Cell Progenitors in Blood

Flaat TD, Smith BF — Poster, Scott-Ritchey Research Center, Auburn University, April 2019

Abstract. Mast cells are multi-functional cells involved in histamine-mediated allergic reactions, wound healing, tissue remodeling, and innate immunity. They are strongly associated with the tumor microenvironment as well as pathologies such as anaphylaxis, allergic rhinitis, and infantile asthma. They originate in bone marrow as hematopoietic stem cells and circulate as uncommitted mast cell progenitors (MCp) expressing $CD117^+$, $CD34^+$, $Fc\varepsilon RI^+$, and $CD90^-$ markers before maturing in peripheral tissues. The long-term goal is to sequence mast cell tumors via a patient-specific precision-medicine approach to identify therapeutic targets. Circulating MCp represent only ~0.1-0.5% of nucleated blood cells, making them difficult to isolate. This study developed a method of flow-cytometric analysis and sorting of $CD117^+$, $CD34^+$, $Fc\varepsilon RI^+$, $CD90^-$ progenitors from canine blood using antibody labeling, with MPT-1 (mast cell tumor immortalized line) as a positive control for CD117/FcεRI and negative for CD90, and normal canine fibroblasts (NCF) as a CD90-positive, CD117/FcεRI-negative control. The gating strategy used unstained samples, Ghost Dye Violet viability dye to focus on live cells, and Fluorescence Minus One (FMO) analysis. Results indicate that sorting the population followed by single-cell sequencing is expected to enable transcriptomic differentiation of the mast cell progenitor population.

Evaluation of Two Methods to Collect Canine Melanocytes

Flaat TD, Agarwal P, Nance R, Platton P, Smith BF — Poster, Phi Zeta Research Day, Auburn University & Texas A&M College of Veterinary Medicine, 2019

Abstract. One of the most difficult aspects of employing precision medicine in cancer treatment is determining the best targets within the tumor. While two patients may present with similar tumors of the same tissue at the same developmental stage, they will likely not respond to identical treatment, posing a significant hurdle to effective regimens. Melanoma is a significant tumor of dogs that typically occurs in the oral mucosa; normal melanocytes from canine melanoma patients are needed to compare transcriptomes to patient melanomas using deep sequencing. This study evaluated two methods for isolating pure canine melanocytes from normal skin. Method 1 (differential adhesion via cell culture): punch biopsies from normal skin were incubated overnight in dispase, the epidermis was separated from the dermis, epidermal cells were dispersed with trypsin into a single-cell suspension, and the cells were cultured in Melanocyte Growth Medium M2. Method 2 (plucking of anagen hair follicles for explant culture): 80-100 hairs were plucked, 30-50 anagen hairs with attached follicles selected, washed in growth media, and explanted onto microporous membranes. Method 1 successfully isolated identifiable melanocytes from non-haired oral mucosa but did not yield pure cultures from haired skin; Method 2 did not yield identifiable melanocytes from hair follicles, as follicles anchor the epidermis in normal canine skin and complicate separation from dermis. Alternative approaches being explored include melanocyte-specific antibodies for flow-cytometric sorting and single-cell sequencing of tumor margins to distinguish normal from cancerous melanocytes.

Hunting Mast Cell Progenitors in Normal Canine Blood

Flaat TD, Sandey M, Smith BF — Poster, Dept. of Pathobiology, Auburn University College of Veterinary Medicine, 2019

Abstract. Circulating mast cell progenitors (MCp) are a vanishingly rare population in normal canine blood, representing roughly 0.1-0.5% of nucleated cells, which makes them a difficult but valuable comparator for mast cell tumor transcriptomes in precision oncology. This poster presents two complementary MACS/FACS strategies for isolating $CD117^+$ progenitors from whole blood. Peripheral blood mononuclear cells were isolated and stained with Ghost Dye Violet 450 to gate on live single cells, with fluorescence-minus-one (FMO) controls used to set marker gates. Magnetic depletion of $CD90^+$ cells was applied as a pre-sort enrichment step, substantially reducing sort time before fluorescence-activated cell sorting of the $CD117^+$ $Fc\varepsilon RI^+$ $CD90^-$ target phenotype, validated against MPT-1 mast cell tumor cells and normal canine fibroblasts as positive and negative controls. The enriched progenitor population is intended as input for single-cell RNA sequencing, which resolves the progenitor transcriptome without requiring a perfectly pure sort.

The Quest to Isolate Normal Control Cell Types for Skin Tumor Analysis

Flaat TD — Seminar presentation, Auburn University, May 2019

The Physics of Hearing and Cochlear Implants

Taylor D. Flaat — University of South Florida Biophysics Seminar, Spring 2015

Patents: contributor to patented synthetic-virology technology (Humane Genomics). Details available on request.

ADDITIONAL EXPERIENCE

Trivia Host, Scorekeeper & Business Owner — UntriviallyJess — 2024 – Present

Luna Jersey City · 902 Brewing Co. · San Patricios · Hudson Hound

- Own and operate a weekly trivia business across four Jersey City venues: curating and fact-checking quizzes, hosting events of 200+ guests, managing staff, and negotiating client contracts.
- Built a **custom loyalty-leaderboard web application** that ingests each night's scores into a living season standing, assigning every team a Loyalty Score, a tier (Casual → Intermediate → Pro), and weekly trend indicators.
- Designed alternative standings modes (Handicap Cup, Giant Slayer Cup, Median Battles) so that dominant teams do not sweep every race, giving newer and casual teams their own competitive ladders and measurably improving repeat attendance.
- Drive crowd size and venue sales through promotion, engaging hosting, and close partnership with restaurant owners; handle scorekeeping, real-time fact-checking, staff onboarding, event-day management, and new-client strategy on a tight weekly schedule.